ML Model Deployment Strategies and Model Serving using FastAPI

Observations Report — Titanic Dataset

Model evaluation: Logistic Regression pipeline, stratified 80/20 split, 712 training rows and 179 testing rows. Test accuracy for the recorded run: **80.45%**.

1. How does Online Inference differ from Batch Inference?

Online inference predicts one input record when a request arrives and returns a result immediately. Batch inference predicts a group of records together, which is useful for offline jobs and can be more efficient per record.

2. What is the difference between Streaming Inference and Batch Inference?

Streaming inference processes each record as it arrives, usually with low latency and continuous processing. Batch inference collects a group of records and processes them together.

3. Why is FastAPI used for model serving?

FastAPI exposes a trained model through HTTP REST endpoints. Pydantic validates incoming request data, and Swagger UI provides interactive API documentation for testing.

4. Why should the same trained model be reused for all inference strategies?

Reusing the same saved pipeline keeps preprocessing and model parameters consistent, makes predictions comparable, and avoids retraining for each strategy. This mirrors deployment where training is separate from inference.

Conclusion

The model was trained once, evaluated on a held-out test split, saved as model.pkl, and loaded for online, batch, streaming, and API inference.